

- 2 (a) Net force on body proportional to rate of change of momentum of the body
(allow eqn with symbols if defined?)

(b) (i) $\Delta p = 140 \times 10^{-3} \times [(5.5 - (-4.0))] = 1.33 \text{ kg m s}^{-1}$

(ii) $F_{\text{net}} = 1.33 / 0.04$
 $= 33.3 \text{ N}$

Average force on ball due to bar

$$= F_{\text{net}} + W_{\text{ball}}$$

$$= (1.33 / 0.04) + (9.81 \times 140 \times 10^{-3})$$

$$= 34.6 \text{ N}$$

By $N_3 \text{ } F_{\text{bar}} = F_{\text{ball}}$.

$$= 34.6 \text{ N} = 35 \text{ N}$$

(iii) Taking moments about B

$$(35 \times 0.75) + (0.450 \times 9.81 \times 0.25) = F_A \times 0.20$$
$$F_A = 137 \text{ N}$$

(iv) $\text{loss} = \frac{1}{2} (0.140) [5.5^2 - 4.0^2] = 0.998 \text{ J}$

(c)

Since initial and final velocities remain the same, the change in momentum of the ball on hitting the bar is unchanged.

However the duration of contact increases, average contact force on bar at point D decreases, leading to lower average contact force at D
And hence a lower exerted force at A